WEIKAI LI

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EDUCATION

UCLA, Los Angeles, California (Ph.D. in Computer Science) 09/2023 – present

- GPA: 4.00/4.00.
- PhD advisor: Prof. Yizhou Sun (and closely collaborating with Prof. Jason Cong).
- Research areas: Graph Neural Network (GNN), the interpretability of large language models (LLMs), AI for Electronic Design Automation (EDA).

Tsinghua University, Beijing, China (B.Eng, Computer Science and Technology) 08/2019 – 06/2023

- GPA: 3.92/4.00 (ranking: 10/195).
- Outstanding Graduate of Tsinghua University (top 10%).
- Undergraduate advisor: Prof. Jie Tang, Prof. Yuxiao Dong, and Prof. Juanzi Li.
- Research areas: Graph Neural Network, LLM's evaluation.

RESEARCH EXPERIENCE

Graduate Student Researcher at UCLA 09/2023 – present
Advised by Prof. Yizhou Sun; closely collaborating with Prof. Jason Cong

- AI for EDA's generalizable model design: enhanced domain generalizability of graph neural networks (GNNs) by Mixture-of-Experts (MoE) in the High-Level Synthesis (HLS) prediction task.

Undergraduate Student Researcher at Tsinghua University 01/2023 – 07/2023
Advised by Prof. Juanzi Li

- LLM's evaluation: analyzed the limitations of LLM's in-context learning on specification-heavy tasks.

Undergraduate Research Intern at UCLA 07/2022 – 12/2022
Advised by Prof. Yizhou Sun

- GNN's approximation: designed an efficient approximation algorithm to calculate the influence of removing a node on a GNN model's prediction, leading to faster detection of influential nodes.

Undergraduate Student Researcher at Tsinghua University 08/2021 – 07/2022
Advised by Prof. Jie Tang and Prof. Yuxiao Dong

- GNN: proposed a new GNN benchmark to improve the evaluation stability and avoid over-tuning hyper-parameters; used mixture-of-experts (MoE) to improve GNN's performance on complex datasets.

PUBLICATIONS

Hierarchical Mixture of Experts: Generalizable Learning for High-Level Synthesis. **Weikai Li**, Ding Wang, Zijian Ding, Atefeh Sohrabizadeh, Zongyue Qin, Jason Cong, Yizhou Sun. AAAI 2025.

Efficient Task Transfer for HLS DSE. Zijian Ding, Atefeh Sohrabizadeh, **Weikai Li**, Zongyue Qin, Yizhou Sun, Jason Cong. ICCAD 2024.

Learning to Compare Hardware Designs for High-Level Synthesis. Yunsheng Bai, Atefeh Sohrabizadeh, Zijian Ding, Rongjian Liang, **Weikai Li**, Ding Wang, Haoxing Ren, Yizhou Sun, Jason Cong. MLCAD 2024.

Fast Inference of Removal-Based Node Influence. **Weikai Li**, Zhiping Xiao, Xiao Luo, Yizhou Sun. WWW 2024.

KoLA: Carefully Benchmarking World Knowledge of Large Language Models. Jifan Yu*, Xiaozhi Wang*, Shangqing Tu, Shulin Cao, Daniel Zhang-Li, Xin Lv, Hao Peng, Zijun Yao, Xiaohan Zhang, Hanming Li, Chunyang Li, Zheyuan Zhang, Yushi Bai, Yantao Liu, Amy Xin, Nianyi Lin, Kaifeng Yun, Linlu Gong, Jianhui Chen, Zhili Wu, Yunjia Qi, **Weikai Li**, Yong Guan, Kaisheng Zeng, Ji Qi, Hailong Jin, Jinxin Liu, Yu Gu, Yuan Yao, Ning Ding, Lei Hou, Zhiyuan Liu, Bin Xu, Jie Tang, Juanzi Li. ICLR 2024.

When does In-context Learning Fall Short and Why? A Study on Specification-Heavy Tasks. Hao Peng*, Xiaozhi Wang*, Jianhui Chen*, **Weikai Li**, Yunjia Qi, Zimu Wang, Zhili Wu, Kaisheng Zeng, Bin Xu, Lei Hou, Juanzi Li. ArXiv preprint 2023.

Rethinking the Setting of Semi-supervised Learning on Graphs. Ziang Li*, Ming Ding*, **Weikai Li**, Zihan Wang, Ziyu Zeng, Yukuo Cen, and Jie Tang. IJCAI 2022.

ONGOING PROJECTS

- **Compositional high-level synthesis prediction:** designing compositional ML models for the high-level synthesis prediction, enhancing the performance and generalizability.
- **Explaining the instruction tuning of LLMs:** interpreting the mechanisms of instruction tuning LLMs, leading to more effective and efficient LLM alignment methods.
- **Verilog code generation:** collecting Verilog code datasets and designing methods to enhance LLM's ability to generate Verilog codes.

SELECTED AWARDS AND HONORS

- NSF Student Travel Award (The Web Conference 2024) 2024
- Outstanding Graduate of Tsinghua University (top 10%) 2023
- Comprehensive Excellence Scholarship for junior year (top 10%) 2022
- Comprehensive Excellence Scholarship for sophomore year (top 10%) 2021
- Academic Excellence Scholarship for freshman year (top 15%) 2020
- Runner-up in Tsinghua Innovation Competition, industry and technology track, freshman group 2019
- First Prize in the National High School Mathematics League 2018

ACADEMIC SERVICES

- Conference reviewer: ICLR 2024, ICML 2024, ACMMM 2024, NeurIPS 2024, LoG 2024, AAAI 2025, ICLR 2025, WWW 2025, ICML 2025.
- Journal reviewer: IEEE TBD 2024, PLOS ONE 2024.
- Conference volunteer: NSF AI for EDA Workshop 2024.

TEACHING EXPERIENCE

- 2024 winter: TA of CS 31 (Introduction to Computer Science I), UCLA.
- 2024 fall: TA of CS 31 (Introduction to Computer Science I), UCLA.
- 2024 summer: TA of CS 97 (Introduction to Data Science), UCLA summer school for high school students.

SKILLS AND INTERESTS

Coding skills

- Programming languages: C, C++, Python, Java.
- Hardware description languages: VHDL, Verilog.
- Others: MatLab, JavaScript.

Interests

- Sports: badminton and table tennis.
- Singing: have been a member of Tsinghua University's choir and UCLA's choir.